Yostina Abera Gobie

+251-925-623-317 | aberayostina@gmail.com | linkedin.com/in/yostina-abera | github.com/BytePhilosopher

Computer Science student who has built ML training pipelines, transactional backend systems, and data pipelines, and contributed to a 50M-user national platform. **400+ algorithmic problems solved**; works in Python and Java.

Education

Addis Ababa University — Expected 06/28
BSc, Computer Science — GPA: 3.91/4.00 — *Addis Ababa, Ethiopia*
- **Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Databases, Operating Systems, Distributed Systems, Machine Learning.

A2SV (Africa to Silicon Valley) — 01/26 – Present
Competitive Programming & Data Structures — *Addis Ababa, Ethiopia*
- Solved **400+ problems** on LeetCode, Codeforces, and HackerRank in a selective, Google-backed program covering graphs, dynamic programming, complexity analysis, and mock interviews.

Experience

iCog Labs — 07/26 – 10/26
Software Engineer Intern — *Addis Ababa, Ethiopia*
- Owned the **baseline residual-learning layer** for a **frozen LLM** (GPT-2) in **JAX**, letting the model adapt to new legal domains **without modifying base weights** while tracking catastrophic forgetting.
- Accelerated training by **JIT-compiling** train and eval steps into fused **XLA** kernels, and made long runs **crash-safe** with atomic checkpoints and load-time shape validation.
- Designed a **Hopfield-memory retrieval benchmark** sweeping the inverse temperature (β) across seeds to pinpoint where recall shifts from blended to exact memory retrieval.

Addis Ababa University — Football Analytics — 11/25 – 06/26
Software / ML Engineering Intern — *Addis Ababa, Ethiopia*
- Cut post-match processing from **O(n·k) to O(n)** with a single hash-map pass and a bounded-deque rolling window, giving **O(1)** ball-velocity updates across full 90-minute matches.
- Built a **Python** pipeline turning broadcast footage into player tracking and possession data, combining **YOLOv8**, ByteTrack, and DBSCAN-based team classification.

Information Network Security Administration (INSA) — 08/25 – 09/25
Software Engineer Intern — *Addis Ababa, Ethiopia*
- Contributed to backend **scalability** of Ethiopia's **national digital ID platform (50M+ registered users)** as part of the engineering team, keeping its REST services reliable under national-scale load.
- Built **Python/SQL** pipelines consolidating network data from multiple sources, replacing manual consolidation with repeatable automated runs.

Projects

Fintech Review Analytics | *Python, PostgreSQL, DistilBERT, GitHub Actions* | GitHub — 05/26
- Built an end-to-end pipeline (scrape → clean → sentiment → PostgreSQL) for **1,200 Play Store reviews** of 3 Ethiopian banks, with unit tests in **CI**; surfaced each bank's top complaint across **5 themes**.

Inventory Management System | *Java, JDBC, SQL Server, Swing* | GitHub — 06/26
- Built a layered **Java** application (UI → service → repository) for multi-location stock, with role-based access for admins, storekeepers, and managers and BCrypt-hashed passwords.
- Made every stock movement an **ACID transaction**: a guarded update makes **negative stock impossible**, transfers commit or roll back as one unit, and a reconciliation query checks balances against the ledger.

CP-Focus | *JavaScript, Chrome Extension API* | GitHub — 03/26 – Present
- Published a Chrome extension that blocks AI chatbots during practice on **6 competitive-programming sites**, unlocking AI only after independent work; **20+ users in its first week** on the Chrome Web Store.

FuelQ – GDG Hackathon | *Next.js, TypeScript, PostgreSQL/Supabase, WebSockets* | GitHub — 05/26 – 07/26
- Co-built a platform replacing **3–6 hour fuel-station lines** with **virtual queues**, with Chapa deposits and live queue positions pushed over WebSockets.

Technical Skills

Languages: Python, Java, C++, JavaScript, TypeScript, SQL
Backend & Data: REST APIs, JDBC, Node.js, Express, Next.js, PostgreSQL, MySQL, SQL Server, Redis, WebSockets
ML: JAX, NumPy, Pandas, Scikit-learn, OpenCV, YOLOv8, HuggingFace Transformers
Tools: Git, Linux, Docker, GitHub Actions, pytest, JUnit

Leadership & Awards

- **Datanomics International Scholarship** ($3,000) · **Quantum Computing Summer School**, Great Distinction · Founded a **1,000+ member** engineering community.